D.C. Is The Clear State-Level Outlier And It Raises The Recent National Rate

School Shootings Capstone Project

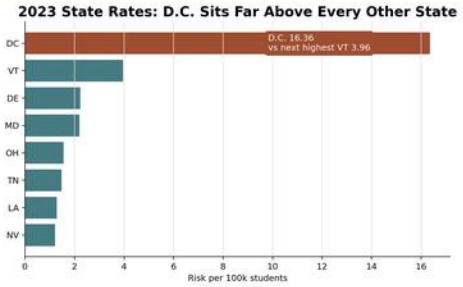
16.36
D.C. risk per 100k in 2023

3.96
Next-highest state risk per 100k in 2023

0.025
D.C. lift to the 2023 national rate

3.38%
Percent lift to the 2023 national rate

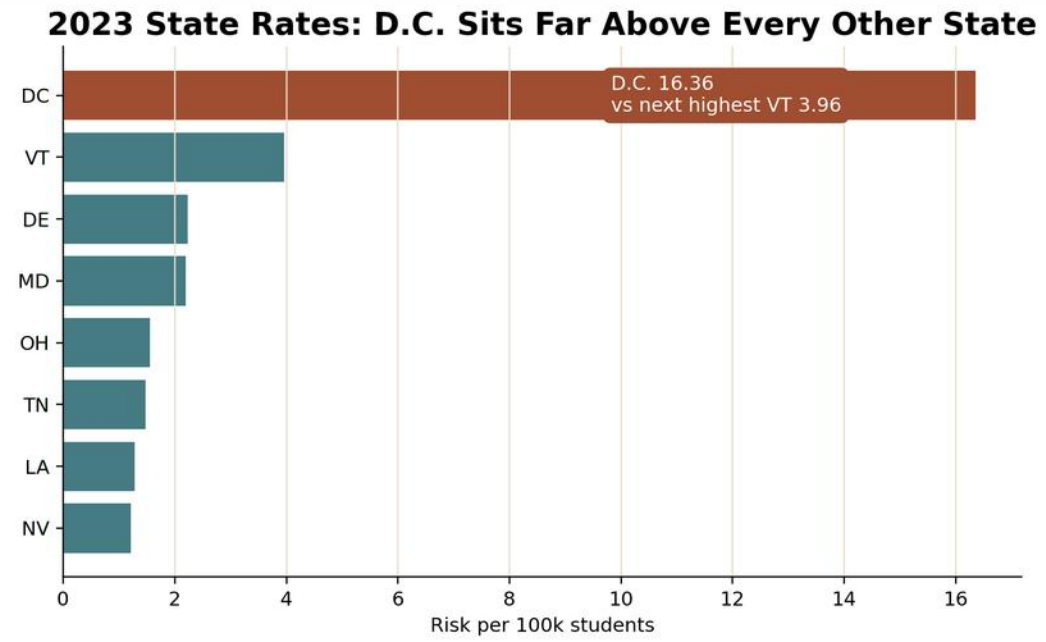
- D.C. is not close to the rest of the state distribution.
- In 2023, D.C. was 16.36 per 100k; the next-highest state was 3.96.
- At the national level, removing D.C. lowers 2023 from 0.774 to 0.749, and 2025 from 0.329 to 0.323.



D.C. Sits Far Above Every Other State In 2023

State-level comparison from the state-year panel used in the notebook.

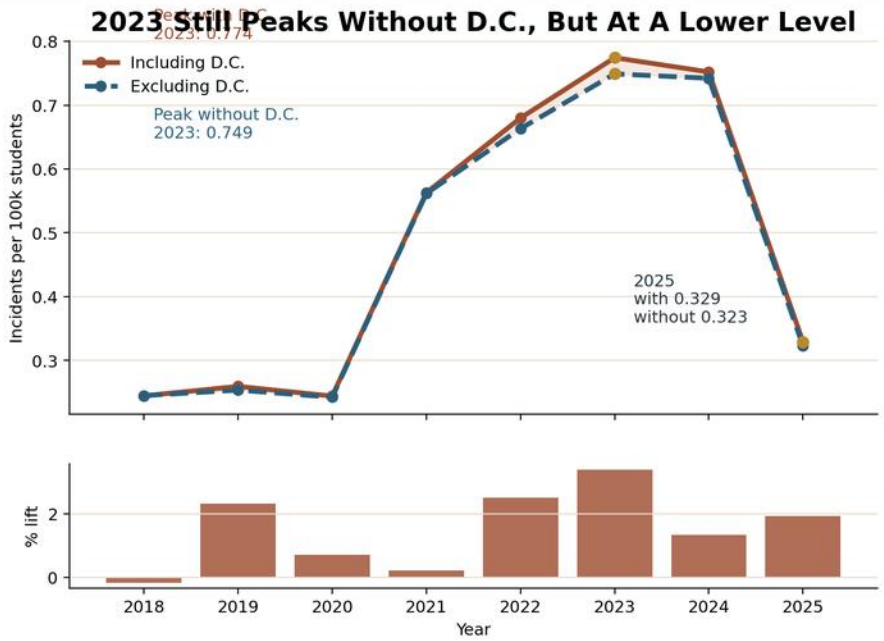
- D.C. is the highest-risk state in the panel by a wide margin in 2023.
- That outlier status comes from a small enrollment denominator combined with a burst in incidents.
- This is why D.C. lifts the national series even though it is only one jurisdiction.



Removing D.C. Lowers The National Rate, But 2023 Still Remains The Peak Year

National series including and excluding D.C., shown directly.

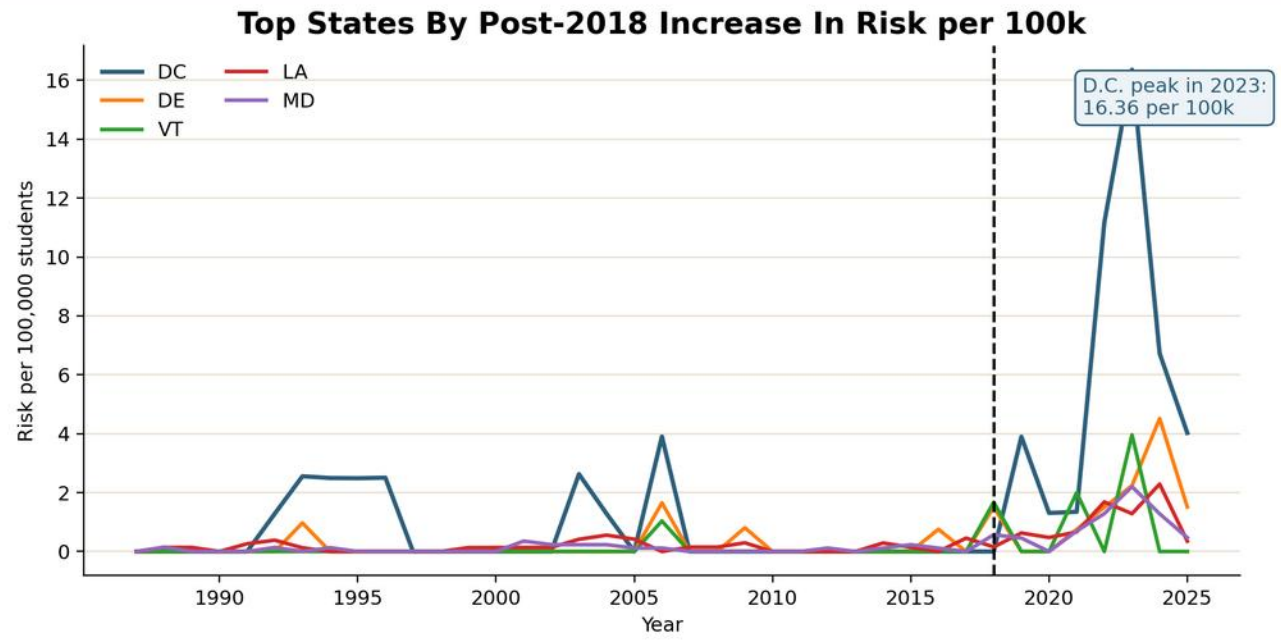
- Peak with D.C.: 2023, 0.774 per 100k. Peak without D.C.: 2023, 0.749 per 100k.
- 2025 with D.C.: 0.329. 2025 without D.C.: 0.323.
- In 2023, D.C. changes the national rate by 0.025 per 100k, or 3.38%.
- Post-2018 average risk falls from 0.481 to 0.473 when D.C. is removed.



D.C. Looks Different Because A Small Denominator Magnifies Short Bursts

The outlier story is real, but the national trend does not depend on it.

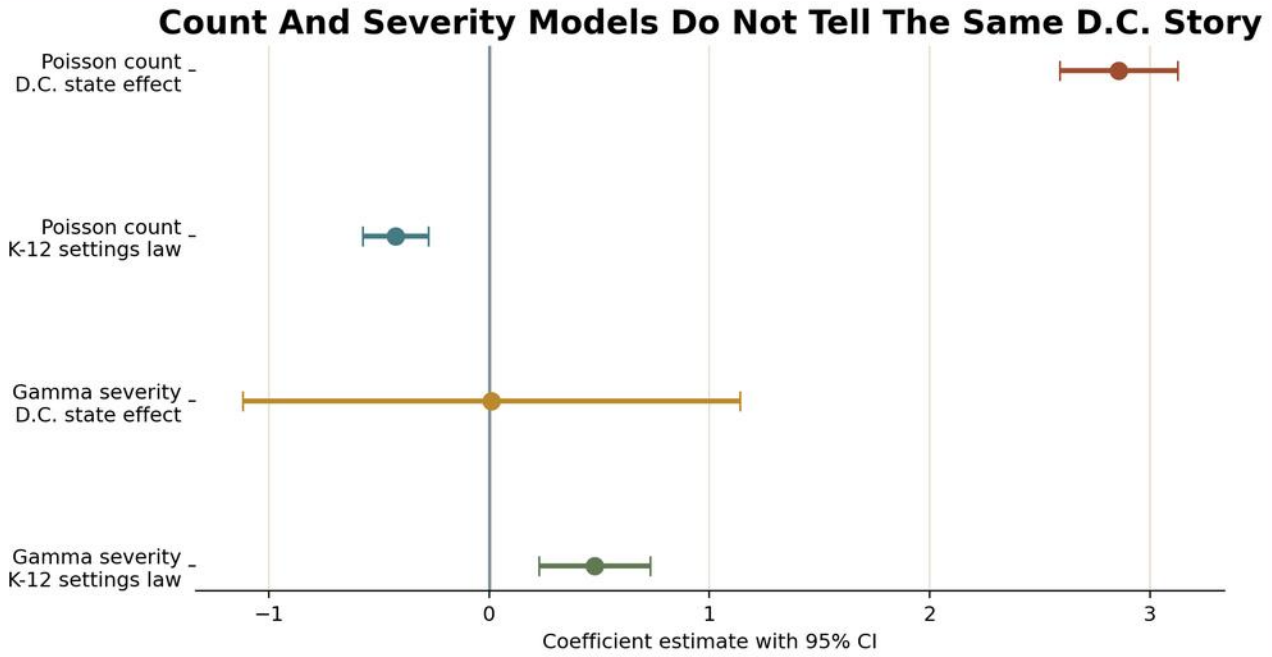
- In 2023, D.C. had 12 incidents across 73,364 students, producing 16.36 per 100k.
- D.C. accounts for 3.43% of 2023 incidents but only 0.16% of students in the panel, which is why its rate spikes so sharply.
- The corrected visual now places D.C. against the other highest-increase states over time, matching the intended notebook-style comparison.
- D.C.'s 2023 rate is 21.8x the national rate excluding D.C., so it is a genuine outlier even though it does not create the national peak year.



Repo Model Outputs Show A D.C. Count Effect, Not A D.C. Severity Effect

These are the notebook-backed model estimates, reconciled with the corrected D.C. sensitivity check.

- README reports NB2 alpha ≈ 0.03 and unstable convergence, so Poisson with clustered SE is the repo's preferred count model.
- Preferred Poisson count model: `C(State)[T.DC] = 2.858` with 95% CI [2.589, 3.126].
- Gamma severity model: `C(State)[T.DC] = 0.011` with 95% CI [-1.118, 1.140], which crosses zero.
- Gamma severity model: `k12_settings_law = 0.479` with 95% CI [0.228, 0.730].



What The Corrected Reading Supports And What It Does Not

The data trend, the D.C. comparison, and the model outputs point to a narrower and more defensible story.

Supported

The national rise after 2018 remains visible and peaks in 2023 even when D.C. is excluded.

D.C. is a real rate outlier in the count data because its student base is small relative to short bursts in incidents.

Not Supported

It is not supported to say D.C. alone creates the national post-2018 trend.

It is not supported to treat D.C. as a severity outlier after conditioning on incidents.

Interpretation

Trend first: the strongest empirical pattern is the national rate increase.

D.C. second: it sharpens the level but does not change the peak year or remove the trend when excluded.

Model third: count and severity models should be discussed separately.

Final Corrected Takeaway: D.C. Amplifies The National Pattern But Does Not Drive It

The corrected conclusion is narrower, data-checked, and presentation-ready.

The state-year panel shows a national risk peak in 2023 with D.C. (0.774) and without D.C. (0.749). D.C. remains an outlier because 12 incidents over 73,364 students produce a very high local rate, but removing D.C. changes the 2023 national rate by only 0.025 per 100k (3.38%). The count model reinforces D.C. as a rate outlier; the severity model does not. The strongest supported claim is the broader post-2018 rise in national risk, not a D.C.-only story.

0.749
2023 national rate excluding D.C.

0.025
D.C. lift to the 2023 national rate

3.38%
Percent effect of D.C. on the 2023 national rate